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Processes for producing orthopedic implants having a subsurface level silicon nitride layer applied via bombardment

Abstract

The process for producing an orthopedic implant having an integrated silicon nitride surface layer includes steps for positioning the orthopedic implant inside a vacuum chamber, mixing nitrogen gas and vaporized silicon atoms in the vacuum chamber, emitting a relatively high energy beam into the mixture of nitrogen gas and vaporized silicon atoms in the vacuum chamber to cause a gas-phase reaction between the nitrogen gas and the vaporized silicon atoms to form reacted precipitate silicon nitride molecules, and driving the precipitate silicon nitride molecules with the same beam into an outer surface of the orthopedic implant at a relatively high energy such that the precipitate silicon nitride molecules implant therein and form at least a part of the molecular structure of the outer surface of the orthopedic implant, thereby forming the integrated silicon nitride surface layer.

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Background/Summary

BACKGROUND OF THE INVENTION

(1) The present invention generally relates to processes for producing orthopedic implants (e.g., hip, knee, shoulder replacements, etc.) having a subsurface level silicon nitride embedded layer applied via ion bombardment, and related implant products. More specifically, the present invention relates to using an ion beam to promote gas-phase reactions for subsequent relatively uniform layered implantation of reacted precipitate silicon nitride molecules into the subsurface of one or more target orthopedic implants.

(2) Orthopedic implants (e.g., prosthetic joints to replace damaged hips, knees, shoulders, etc.) are commonly made of metal alloys such as cobalt chromium (CoCr) or titanium (Ti-6Al-4V). The mechanical properties of such metal alloys are particularly desirable for use in load-bearing applications, such as orthopedic implants. Although, when orthopedic implants are placed within the body, the physiological environment can cause the implant material to wear and corrode over time (especially articulatory surfaces), sometimes resulting in complications that require revision surgery. While hip and knee replacement surgery has been reported to be successful at reducing joint pain for 90-95% of patients, there are several complications that remain and the potential for

revision surgery increases at a rate around 1% per year following a successful surgery. These complications can include infection and inflammatory tissue responses stemming from tribological debris particles from metal alloy implants, such as cobalt chromium, as a result of wear and corrosion over time.

(3) To reduce the risk of complications from orthopedic implants, ceramic coatings have been applied to address the coefficient of friction of a wear couple, to specifically improve the surface roughness, and to reduce adhesion of a broad range of bacteria for purposes of reducing the rate of infection. For example, alumina (Al_2O_3) and zirconia (ZrO_2) are ceramics that have been used to coat the surfaces of orthopedic implants. These ceramic materials provide high wear resistance, reduced surface roughness, and high biocompatibility. But, both materials are not optimal for the fatigue loading of non-spherical geometry of most orthopedic implants due to poor tensile strength and low toughness. Accordingly, the disadvantages of these ceramic coatings, while addressing issues related to high wear resistance and surface roughness, cannot address other failure modes such as tensile strength and impact stresses.

(4) Conventionally, ceramic coatings such as silicon nitride have been applied to the implant surface by a chemical vapor deposition (CVD) process or a physical vapor deposition (PVD) process. In one example, a PVD process is used to coat an implant joint with an external layer of silicon nitride. More specifically, such a process includes placing the implant, a silicon-containing material, and nitrogen gas (N_2) in a chamber that is heated to between 100-600 degrees Celsius. In response to the high temperatures, silicon atoms sputter from the silicon-containing material and subsequently react with the nitrogen gas at the heated surface of the implant to deposit a silicon nitride over-coat. One problem with this process is that there is no diffusion of the deposited silicon nitride molecules into the substrate material. That is, the silicon nitride is simply applied as an over-surface coating having a distinct boundary line between the deposited over-coating and the underlying substrate of the orthopedic implant. The adverse result is that the silicon nitride still experiences relatively poor surface adhesion and, over time, this over-surface coating can wear off, especially when the surface is an articulating surface (e.g., a ball-and-socket joint).

(5) While vapor deposition of silicon nitride has been shown to work as an over-surface coating to certain orthopedic materials, such application is typically more expensive and less efficient than alumina or zirconia ceramic coatings. Moreover, it is often difficult, if not impossible, to attain a uniform application of silicon nitride to all surfaces of the orthopedic implant using known vapor deposition processes, such as those mentioned above. As a result, some areas of the over-surface coating have an undesirably thin layer of silicon nitride, wherein such areas are even more prone to reduced protection and wear. Alternatively, silicon nitride has also been used as the bulk or base material for orthopedic implants, but the production of a silicon nitride-based orthopedic implant is limited in size and inefficient to produce.

(6) Recently, newer coating processes have been developed to provide greater adhesion by promoting diffusion of the coating material at the interface of the substrate and coating layers. Ion beam enhanced deposition (IBED), also known as ion beam assisted deposition (IBAD), is a process by which accelerated ions drive a vapor phase coating material into the subsurface of a substrate. Coatings applied by IBED may have greater adhesion than similar coatings applied by a conventional PVD process. Coatings applied by IBED may also have less delamination under impact stresses. For example, U.S. Pat. No. 7,790,216 to Popoola et al., the contents of which are herein incorporated by reference in their entirety, discloses a method of bombarding a medical implant with zirconium ions and then heating the implant in an oxygenated environment to induce the formation of zirconia (ZrO_2) at the surface. In this respect, the ion beam drives the zirconium ions to a certain depth within the surface of the implant known as the “intermix zone”. Heat treatment within the oxygenated environment results in an embedded zirconia surface layer of approximately 5 micrometer (μm) thickness. The zirconia surface layer effectively penetrates the substrate and thereby resists delamination. But, this production method can be inefficient due to the

high energy requirement for the heat treatment step. Likewise, the mechanical properties of the zirconia surface layer formed are not as desirable as those of a silicon nitride surface layer, which is incompatible with a heat treatment step.

(7) There exists, therefore, a need in the art for processes for producing orthopedic implants having a subsurface silicon nitride layer applied via ion bombardment that provides greater integration of silicon nitride into the implant, thereby providing greater resistance to the emission of tribological debris. Such processes may include placing an orthopedic implant in a vacuum chamber, diffusing gaseous nitrogen and silicon within the chamber, and bombarding a surface of the orthopedic implant with an ion beam sufficient to promote gas-phase reactions to form silicon nitride and drive reacted precipitate silicon nitride molecules into the subsurface of the medical implant. The present invention fulfills these needs and provides further related advantages.

SUMMARY OF THE INVENTION

(8) In one embodiment, a process for producing an orthopedic implant having an integrated silicon nitride surface layer as disclosed herein may include steps for positioning the orthopedic implant inside a vacuum chamber, mixing nitrogen gas and vaporized silicon atoms in the vacuum chamber, emitting a relatively high energy beam into the mixture of nitrogen gas and vaporized silicon atoms in the vacuum chamber to cause a gas-phase reaction between the nitrogen gas and the vaporized silicon atoms to form reacted precipitate silicon nitride molecules, and driving the precipitate silicon nitride molecules with the same beam into an outer surface of the orthopedic implant at a relatively high energy such that the precipitate silicon nitride molecules implant therein and form at least a part of the molecular structure of the outer surface of the orthopedic implant, thereby forming the integrated silicon nitride surface layer. An intermix layer may be formed underneath the integrated silicon nitride surface layer, depending on the energy intensity of the beam. Here, the intermix layer may include a mixture of the precipitate silicon nitride molecules and a base material of the orthopedic implant.

(9) In one aspect of this embodiment, the beam may include an ion beam that emits nitrogen ions selected from the group consisting of N^+ ions or N_2^+ ions. Accordingly, the emitting step may include delivering the nitrogen ions at a rate of about 1-5 nitrogen ions for each vaporized silicon atom. In this embodiment, when the rate is about one nitrogen ion for each silicon atom, the integrated silicon nitride surface layer may include $Si_{3.3}N_{4.4}$. Alternatively, when the rate is about five nitrogen ions for each silicon atom, the integrated silicon nitride surface layer may include $SiN_{0.3}$. During the driving step, the beam may also include an energy level greater than 102 kiloelectron volts (KeV), yet the temperature of the outer surface of the orthopedic implant may be below 200 degrees Celsius. The beam may propagate relative to the orthopedic implant, or the positioning step may include mounting the orthopedic implant to a selectively movable platen for repositioning an orientation of the orthopedic implant relative to the beam.

(10) In other aspects of this embodiment, the outer surface of the orthopedic implant may be cleaned prior to or during implantation by setting the beam to an energy level between about 1-1000 electron volts. Additionally, an evaporator positioned within the vacuum chamber may vaporize silicon atoms off a silicon ingot at a rate determined by the desired ratio of nitrogen molecules to silicon atoms inside the vacuum chamber at any given time during the process. Here, for example, the formation rate of the precipitate silicon nitride molecules may be regulated by adjusting the beam energy or beam density. Additionally, the quantity of nitrogen gas and/or the quantity of vaporized silicon atoms may be further controlled by backfilling the vacuum chamber with the same. The resultant integrated silicon nitride surface layer may have a substantially uniform thickness where the silicon nitride molecules are driven into the orthopedic implant. In some embodiments, the driving step may include the step of applying the integrated silicon nitride surface layer to less than an entire outer surface area of the orthopedic implant. The integrated silicon nitride surface layer may substantially include the silicon nitride molecules.

(11) In another process disclosed herein, producing an orthopedic implant having an integrated

silicon nitride surface layer may include steps for positioning the orthopedic implant inside a vacuum chamber, vaporizing silicon atoms off a silicon ingot with an evaporator, mixing nitrogen gas and the vaporized silicon atoms in the vacuum chamber, and emitting a relatively high energy beam having an energy level greater than 102 kiloelectron volts (KeV) into the mixture of nitrogen gas and vaporized silicon atoms in the vacuum chamber to cause a gas-phase reaction between the nitrogen gas and the vaporized silicon atoms, thereby forming reacted precipitate silicon nitride molecules. The outer surface of the orthopedic implant may be cleaned with the beam by setting the initial energy level between about 1-1000 electron volts. Thereafter, the precipitate silicon nitride molecules may be driven with the same beam into the outer surface of the orthopedic implant albeit at the same or a relatively higher energy level such that the precipitate silicon nitride molecules implant therein and form at least a part of the molecular structure of the outer surface of the orthopedic implant simultaneously while maintaining the outer surface of the orthopedic implant at a temperature below 200 degrees Celsius. Such a process may form the integrated silicon nitride surface layer therein.

(12) The orthopedic implant may be mounted to a selectively movable platen within the vacuum chamber for repositioning an orientation of the orthopedic implant relative to the beam. In this embodiment, the formation rate of the precipitate silicon nitride molecules may be regulated by adjusting the beam energy or beam density. The driving step may also include the step of applying the integrated silicon nitride surface layer to less than an entire outer surface area of the orthopedic implant. Additionally, backfilling the vacuum chamber with the nitrogen gas or the vaporized silicon atoms may maintain the desired ratios, i.e., in an embodiment where the beam includes an ion beam including nitrogen ions selected from the group consisting of N^+ ions or N_2^+ ions, the emitting step may include the step of delivering the nitrogen ions at a rate of about 1-5 nitrogen ions for each vaporized silicon atom. When the rate is about one nitrogen ion for each silicon atom, the integrated silicon nitride surface layer may include Si_3N_4 , and, when the rate is about five nitrogen ions for each silicon atom, the integrated silicon nitride surface layer may include SiN_3 .

(13) In another aspect of this embodiment, an intermix layer may be formed underneath the integrated silicon nitride surface layer. Here, the intermix layer may include a mixture of the precipitate silicon nitride molecules and a base material of the orthopedic implant. The integrated silicon nitride surface layer may include a substantially uniform thickness where driven into the orthopedic implant, such as by a propagating the beam, and the integrated silicon nitride surface layer may substantially include the silicon nitride molecules.

(14) In another embodiment, an orthopedic implant as disclosed herein may include or be made from a base material that includes a metal alloy selected from the group consisting of cobalt, titanium, and zirconium, a ceramic material selected from the group consisting of alumina (Al_2O_3) and zirconia (ZrO_2), an organic polymer, or a composite organic polymer. The orthopedic implant may also include an intermix layer that includes a mixture of the base material and a plurality of silicon-based molecules (e.g., Si_3N_4 or SiN_3) implanted into the base material. The intermix layer may include a thickness of about 0.1-100 nanometers and include a subsurface level of silicon-based molecules.

(15) Additionally, the orthopedic implant may include an integrated silicon nitride surface layer forming at least part of a molecular structure of an outer surface of the orthopedic implant. Here, the integrated silicon nitride surface layer and the base material may cooperate with one another to sandwich the intermix layer in between. In one embodiment, the integrated silicon nitride surface layer may have a relatively uniform depth. In another embodiment, the integrated silicon nitride surface layer may cover less than an entire surface area of the base material. The base material may include the integrated silicon nitride surface layer on an articulating surface only and the orthopedic implant may have an electrical resistivity of about $10^{16} \Omega \cdot cm$ when incorporating the integrated silicon nitride surface layer therein.

(16) In other aspects of the embodiments disclosed herein, the silicon nitride surface layer and the base material may form an alloy bond therebetween at an atomic level by ion bombardment. The intermix layer and the integrated silicon nitride surface layer may have an aggregate thickness of about 1-10,000 nanometers and the orthopedic implant may include a hip implant, a knee implant, or a shoulder implant.

(17) In another embodiment of an orthopedic implant as disclosed herein, the orthopedic implant may generally be made from a base material that includes a metal alloy selected from the group consisting of cobalt, titanium, and zirconium, a ceramic material selected from the group consisting of alumina (Al_2O_3) and zirconia (ZrO_2), an organic polymer, or a composite organic polymer. An intermix layer having a thickness of about 0.1-100 nanometers may include a mixture of the base material and a plurality of subsurface level silicon-based molecules implanted into the base material. The silicon-based molecules may include Si_3N_4 or SiN , and the integrated silicon nitride surface layer may cover less than an entire surface area of the base material. Additionally, the orthopedic implant may include an integrated silicon nitride surface layer having a relatively uniform thickness and forming at least part of the molecular structure of an articulating surface of the orthopedic implant. Here, the integrated silicon nitride surface layer and the base material may cooperate to sandwich the intermix layer in between. The intermix layer and the integrated silicon nitride surface layer may have an aggregate thickness of about 1-10,000 nanometer and the orthopedic implant incorporating the integrated silicon nitride surface layer may have an electrical resistivity of about $10^{16} \Omega \cdot \text{cm}$.

(18) In another aspect, the processes disclosed herein for producing orthopedic implants having a subsurface level silicon nitride layer implanted via ion bombardment may include mounting an orthopedic implant to an angling and/or rotating part platen assembly located inside a vacuum chamber. An ion beam generator may emit a beam of energized ions that bombard an outer surface of the angling and/or rotating orthopedic implant to cleanse the surface of contaminants. The chamber may then be filled with a gaseous mixture of vaporized silicon atoms and nitrogen gas molecules. The ion beam excites the gaseous silicon atoms and nitrogen molecules to promote a reaction that forms silicon nitride. The ion beam subsequently drives the precipitated silicon nitride molecules into the angling and/or rotating surface of the orthopedic implant to form an ion bombarded subsurface layer of silicon nitride, the depth of which depends on the intensity of the ion beam driving the silicon nitride molecules into the subsurface of the orthopedic implant.

(19) Other features and advantages of the present invention will become apparent from the following more detailed description, when taken in conjunction with the accompanying drawings, which illustrate, by way of example, the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings illustrate the invention. In such drawings:

(2) FIG. 1 is a flowchart illustrating a process for producing orthopedic implants having a subsurface level silicon nitride bombardment layer, as disclosed herein;

(3) FIG. 2 is a diagrammatic view of an ion beam enhanced deposition (IBED) chamber, in accordance with the embodiments disclosed herein;

(4) FIG. 3a is a diagrammatic view illustrating interaction of an ion beam with a gaseous mixture of vaporized silicon atoms and nitrogen gas molecules;

(5) FIG. 3b is a diagrammatic view illustrating the ion beam promoting reaction of the gaseous mixture of the vaporized silicon atoms and the nitrogen gas molecules to form gaseous silicon nitride molecules;

(6) FIG. 4a is a diagrammatic view illustrating the ion beam driving the gaseous silicon nitride

molecules into the angling and/or rotating surface of the orthopedic implant, thereby forming a subsurface intermixed layer;

(7) FIG. 4b is a diagrammatic view illustrating the ion beam further driving the gaseous silicon nitride molecules into the angling and/or rotating surface of the orthopedic implant, thereby forming a subsurface silicon nitride layer of relatively uniform thickness over the subsurface intermixed layer; and

(8) FIG. 5 is a cross-sectional view of the orthopedic implant having the subsurface silicon nitride layer produced by the ion beam implantation or bombardment of the gaseous silicon nitride molecules therein.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(9) As shown in the exemplary drawings for purposes of illustration, the processes for producing orthopedic implants having a subsurface level silicon nitride bombardment layer is referred to by numeral (100) with respect to the flowchart in FIG. 1, while FIGS. 2-4b more specifically illustrate the operation of said processes, and FIG. 5 illustrates an exemplary orthopedic implant with a subsurface level silicon nitride bombardment layer 10. More specifically, the first step (102) in the process (100), as shown in FIG. 1, is to mount an orthopedic implant workpiece 12 onto an angling and/or rotating part platen 14 inside a vacuum chamber 16 suitable for performing ion beam implantation (e.g., ion beam enhanced deposition (IBED)). The processes disclosed herein improve the integration of silicon nitride into the orthopedic implant by kinetically driving silicon nitride molecules into a subsurface layer of the orthopedic implant. This improved integration of silicon nitride reduces delamination and prevents future wear and corrosion. Furthermore, the processes disclosed herein can reduce energy costs by performing the IBED process at temperatures well below 200 degrees Celsius and without a heat treatment step. Accordingly, the processes disclosed herein also reduce energy costs associated with manufacturing the related implant products.

(10) More specifically, FIG. 2 illustrates the orthopedic implant workpiece 12 mounted to the angling and/or rotating part platen 14 within the vacuum chamber 16. The orthopedic implant workpiece 10 may be made from a variety of metal alloys known in the art, such as cobalt, titanium, zirconium alloy, etc. In other embodiments, the orthopedic implant workpiece 10 may be made from ceramic materials known in the art, such as alumina (Al.sub.2O.sub.3) or zirconia (ZrO.sub.2). In still other embodiments, the orthopedic implant workpiece 10 may be made from organic polymers or composites of organic polymers. Of course, persons of ordinary skill in the art may recognize that the processes disclosed herein may be used with other types of materials, and that the scope of the present disclosure should not be limited only to those materials mentioned above. The part platen 14 may be able to rotate about a center axis 18 and/or tilt about a vertical axis 20 to facilitate maximum exposure of the orthopedic implant workpiece 10 to an ion beam 22 during the silicon nitride implantation process. In one embodiment, the orthopedic implant workpiece 10 may couple to the part platen 14 via an attachment 24 that may include a grip, clamp, or other device having a high friction surface to retain (e.g., by compression fit) the orthopedic implant workpiece 10. In this respect, any attachment known in the art capable of sufficiently securing the orthopedic implant workpiece 10 to the part platen 14, as the part platen 14 rotates and/or tilts, will suffice. The vacuum chamber 16 maintains a high vacuum environment during the silicon nitride implantation process to promote the propagation of ions from the ion beam 22 toward the surfaces of the orthopedic implant workpiece 10. The high vacuum environment additionally reduces the amount of contaminant gases present to prevent contamination of a silicon nitride layer 26 (shown best in FIG. 5) subsequently bombarded or implanted into a surface 28 of the orthopedic implant workpiece 10. In further embodiments, a plurality of the part platens 12 may be present within the vacuum chamber 16 during the silicon nitride implantation process. In this embodiment, a plurality of the orthopedic implant workpieces 10 may be mounted in an array on each of the part platens 12 to produce multiple silicon nitride-implanted orthopedic implants 10 during each silicon nitride implantation process.

(11) Once the orthopedic implant workpiece **10** has been mounted on the part platen **14**, the next step (**104**), as shown in FIG. **1**, is to energize an ion beam generator **30** to produce the ion beam **22** of energized nitrogen ions capable of penetrating into the surface **28** of the orthopedic implant workpiece **10** as it rotates about the center axis **18** and/or pivots about the vertical axis **20**. Here, FIG. **2** illustrates the ion beam generator **30** emitting the ion beam **22** directed at the surface **28** of the orthopedic implant workpiece **10**. In one example, the ion beam generator **30** can include a Kaufman ion source (e.g., a gridded broad beam ion source of permanent magnet design). The ion beam generator **30** can be capable of delivering nitrogen ions (e.g., N^+ ions and/or $N_{sub.2}^{+}$ ions) at beam energies up to 102 kiloelectron volts (KeV) at currents up to 6 mA. The ion beam **22** initially bombards the surface **28** of the orthopedic implant workpiece **10** with energized nitrogen ions during an ion beam cleaning process, thereby cleaning and augmenting the surface **28** of the orthopedic implant workpiece **10**. Specifically, the initial bombardment of the orthopedic implant workpiece **10** during step (**104**) efficiently removes absorbed water vapor, hydrocarbons, and other substrate surface contaminants from the surface **28** of orthopedic implant workpiece **10**. Removal of the substrate surface contaminants results in better implantation when the silicon nitride layer **26** is subsequently added to the subsurface of the orthopedic implant workpiece **10**. Step (**104**) may also create defects in the surface **28** of orthopedic implant workpiece **10** which further promotes the subsequent implantation of the silicon nitride layer **26**. At step (**104**) of the silicon nitride implantation process, relatively low energy ions (e.g., at beam energies between 1-1000 eV) can be employed to minimize sputtering at the surface **28** of orthopedic implant workpiece **10**, while still being sufficiently energetic to produce the desired effects mentioned above.

(12) Once the surface **28** of the orthopedic implant workpiece **10** has been cleaned and augmented by the ion beam **22**, the next step (**106**) in accordance with FIG. **1** is to diffuse a gaseous mixture **32** of nitrogen gas ($N_{sub.2}$) and vaporized silicon (Si) into the vacuum chamber **16**. High purity nitrogen gas and a silicon ingot can be used as source materials to produce the gaseous mixture **32**. In this regard, as shown in FIG. **2**, an evaporator **34** located within the vacuum chamber **16** may produce a quantity of vaporized silicon atoms **36** by electron beam evaporation. Here, the evaporator **34** may direct an electron beam (not shown) at a silicon ingot workpiece (also not shown) to provide a direct flux of the vaporized silicon atoms **36**, which disperse within vacuum chamber **16** as shown. The high purity nitrogen gas molecules **38** may then be subsequently introduced into the vacuum chamber **16** through an inlet **40**. The nitrogen gas molecules **38** then mix with the vaporized silicon atoms **36** to form the gaseous mixture **32**. The ion beam **22** may then energize the resulting gaseous mixture **32** to form silicon nitride molecules **42**, as discussed in detail herein.

(13) Once the gaseous mixture **32** has been introduced into the vacuum chamber **16**, the next step (**108**) as shown in FIG. **1** is to promote and control the reaction of the vaporized silicon atoms **36** and the nitrogen gas molecules **38** in gaseous mixture **32** using the ion beam **22**, as shown in FIGS. **3a-3b**. First, the positively charged nitrogen ions of the ion beam **22** collide with and kinetically excite the vaporized silicon atoms **36** and the nitrogen gas molecules **38** to promote the reaction process generally shown in FIG. **3a**. Once kinetically excited, the vaporized silicon atoms **36** react with the nitrogen gas molecules **38** to form the gaseous silicon nitride molecules **42** as shown in FIG. **3b**. The rate of formation of the gaseous silicon nitride molecules **42** can be controlled by varying the energy and/or the density of the ion beam **22**. For example, increasing the energy and/or density of the ion beam **22** increases the rate of formation of the gaseous silicon nitride molecules **42**, and vice versa. As the vaporized silicon atoms **36** and the nitrogen gas molecules **38** react during step (**108**) to form gaseous silicon nitride molecules **42**, a controlled backfill of vaporized silicon atoms **36** and/or the nitrogen gas molecules **38** may be employed to maintain the desired concentration of reactant molecules in the vacuum chamber **16**.

(14) In some embodiments of the processes disclosed herein, steps (**106**) and (**108**) may be performed without halting the cleaning process described in step (**104**). That is, the vaporized

silicon atoms **36** and the nitrogen gas molecules **38** may be introduced into the vacuum chamber **16** without halting the ion beam cleaning process of step **(104)**. In this way, the ion beam **22** immediately begins promoting the reaction of the vaporized silicon atoms **36** and the nitrogen gas molecules **38** once introduced into vacuum chamber **16**. This can be more efficient from a manufacturing standpoint by reducing the duration required to perform the silicon nitride implantation process disclosed herein. Additionally, introducing the vaporized silicon atoms **36** and the nitrogen gas molecules **38** without halting the cleaning process can prevent subsequent contamination of the substrate surface **28**. This may further promote generation of the subsurface silicon nitride layer **26** in the surface **28** of the orthopedic implant workpiece **10**.

(15) Once the gaseous silicon nitride molecules **42** are formed, the ion beam **22** subsequently drives the silicon nitride molecules **42** into the surface **28** of the rotating and/or pivoting orthopedic implant workpiece **10**, per step **(110)** in FIG. **1**. The high-energy nitrogen ions of the ion beam **22** collide with the silicon nitride molecules **42** to impart kinetic energy thereto. The energized silicon nitride molecules **42** subsequently collide with the surface **28** of the orthopedic implant workpiece **10** and bombard or implant therein, thereby initially forming a subsurface intermixed layer **44**, as shown in FIG. **4a**. The silicon nitride molecules **42** bombarded or implanted therein integrate with the surface **28**, as opposed to simply be deposited on the surface **28** as an over surface coating, as is the current practice with known silicon nitride deposition procedures. The intermixed layer **44** is basically a transition region wherein the surface molecules **46** of the orthopedic implant workpiece **10** become intermixed with the silicon nitride molecules **42** as a result of the energized bombardment by way of the ion beam **22**. The accumulation of silicon nitride molecules **42** within the intermixed layer **44** results in alloyed silicon nitride molecules **42** and substrate molecules **46**. By varying the energy and/or density of the beam **22**, persons skilled in the art can vary the depth into which the silicon nitride molecules **42** are driven.

(16) As the intermixed layer **44** develops, the ion beam **22** continues to drive the silicon nitride molecules **42** into the subsurface of the surface **28** of the orthopedic implant workpiece **10**. As shown in FIG. **4b**, through time, the silicon nitride layer **26** subsequently begins to form above the intermixed layer **44**. The depth the silicon nitride layer **26** forms into the subsurface of the surface **28** varies according to various variables, including the energy and/or density of the ion beam **22** (i.e., higher energy or a greater density results in a thicker or deeper silicon nitride layer **26**, and vice versa) and/or the duration of bombardment with the ion beam **22** (i.e., a longer bombardment in a particular area may result in a thicker or deeper silicon nitride layer **26**, and vice versa). Similarly, varying the rate of nitrogen ion arrival can affect the stoichiometry of the resulting silicon nitride layer **26**. For example, the nitrogen ion arrival rate may be in the range of about one (1) nitrogen ion to about five (5) nitrogen ions for each silicon atom in the gaseous mixture **32**. In some embodiments of the processes disclosed herein, a nitrogen ion arrival rate closer to one (1) ion for each silicon atom can result in the silicon nitride layer **26** that includes Si.sub.3N.sub.4. In other embodiments, a nitrogen ion arrival rate closer to five (5) ions for each silicon atom can result in a silicon nitride layer **26** that includes SiN.sub.3. Persons of ordinary skill in the art may vary the nitrogen ion arrival rate to obtain a silicon nitride phase suitable for the desired application.

(17) As a result of step **(110)**, the silicon nitride layer **26** is molecularly integrated into the subsurface of the surface **28** (e.g., as shown in FIG. **5**) of the orthopedic implant workpiece **10** and exhibits superior retention relative to silicon nitride coatings simply deposited as an over coating on the surface **28** by traditional PVD processes. This is due, at least in part, to the high strength of the alloy bond formed at an atomic level by the ion bombardment, which creates the intermixed layer **44** between the silicon nitride layer **26** and the surface molecules **46** of the orthopedic implant workpiece **10**. As such, this ultimately changes the atomic foundation of the subsurface of the orthopedic implant workpiece **12**. As the bombardment continues, the outermost silicon nitride layer **26** builds up, and does so over the entire orthopedic implant workpiece **12** as it rotates and/or pivots with the part platen **14**. Although, of course, the processes disclosed herein may include

application to only a part of the orthopedic implant workpiece **12**, e.g., the articulation surfaces, as opposed to the entire orthopedic implant workpiece **12**. The articulation surfaces may later be polished, along with adjacent surfaces or other fixation surfaces. The material properties of the orthopedic implant workpiece **12**, in combination with the energy intensity characteristics of the ion beam **22**, limit the penetration depth to attain a more consistently uniform silicon nitride layer **26**. In this regard, the silicon nitride layer **26** is less likely to delaminate from the orthopedic implant workpiece **10** when compared to conventional PVD coatings. As such, the processes and implants disclosed herein are able to attain the benefits of silicon nitride across different types of surface finishes and surface requirements of an orthopedic implant.

(18) During step (**110**), the surface **28** of the orthopedic implant workpiece **10** increases in temperature as a result of bombardment by the ion beam **22**. As such, a cooler can be utilized to cool the silicon nitride layer **26**, the intermixed layer **44**, and/or orthopedic implant workpiece **10** in general to prevent adverse or unexpected changes in the material properties due to heating. In this respect, cooling may occur in and/or around the area of the orthopedic implant workpiece **10** being bombarded or implanted with the silicon nitride layer **26**, and including the part platen **14**. Water or air circulation-based coolers may be used with the processes disclosed herein to provide direct or indirect cooling of the orthopedic implant workpiece **10**.

(19) FIG. 5 is a diagrammatic cross-sectional view illustrating the surface **28** of the orthopedic implant workpiece **10**, including the resultant intermixed layer **44** and the silicon nitride layer **26** formed into the subsurface thereof. The processes disclosed herein result in the intermixed layer **44** having a thickness **48** and the silicon nitride layer **26** having an implantation thickness **50**, as shown in FIG. 5. The intermixed layer **44** is positioned generally between the unaffected surface molecules **46** and the silicon nitride layer **26**. Accordingly, the intermixed layer **44** may form a uniform layer immediately above the unaffected surface molecules **46**, such as designated by a boundary **52**, and the silicon nitride layer **26** may form a uniform layer immediately above the intermixed layer **44**, such as designated by a boundary **54**. The intermixed width **48** and the depth of the boundary **52** may vary depending on the energy and/or density of the ion beam **22**, to increase (i.e., higher energy and/or density) or decrease (i.e., lower energy and/or density) the integration or implantation of the silicon nitride molecules **42** into the subsurface of the surface **28** of the orthopedic implant workpiece **10**. Likewise, the implantation thickness **50** and the depth of the boundary **54** may vary depending on the energy and/or density of the ion beam **22**, to increase (i.e., higher energy and/or density) or decrease (i.e., lower energy and/or density) the integration or implantation of the silicon nitride molecules **42** into the subsurface of the surface **28** of the orthopedic implant workpiece **10**. In an exemplary embodiment, the intermixed width **48** may be between 0.1-100 nanometers, while the implantation thickness **50** may be between 1-10,000 nanometers.

(20) The resulting silicon nitride layer **26** may exhibit excellent tribological properties, including long-term material stability and high biocompatibility, at least relative to alumina. Likewise, silicon nitride is semitransparent to X-rays and is not magnetic, thereby allowing MRI of soft tissues proximal to silicon nitride coated implants. Meanwhile, silicon nitride also has wear rates comparable to alumina. Furthermore, unlike zirconia, which is a good conductor of electricity, silicon nitride advantageously has high electrical resistivity, such as on the order of $10^{16} \Omega \cdot \text{cm}$.

(21) Although several embodiments have been described in detail for purposes of illustration, various modifications may be made without departing from the scope and spirit of the invention. Accordingly, the invention is not to be limited, except as by the appended claims.

Claims

1. A process for producing an orthopedic implant having an integrated silicon nitride surface layer, comprising the steps of: positioning the orthopedic implant inside a vacuum chamber; vaporizing silicon atoms inside the vacuum chamber; emitting ions via an ion beam into the vaporized silicon atoms in the vacuum chamber to cause a collision between the ions and the vaporized silicon atoms to form silicon nitride molecules; driving the silicon nitride molecules with the same ion beam into an outer surface of the orthopedic implant such that the silicon nitride molecules implant therein and form at least a part of a molecular structure of the outer surface of the orthopedic implant simultaneously while maintaining the outer surface of the orthopedic implant at a temperature below 200 degrees Celsius, thereby forming the integrated silicon nitride surface layer; and forming an intermix layer underneath the integrated silicon nitride surface layer, the intermix layer including a mixture of subsurface level silicon nitride molecules and a base material of the orthopedic implant, wherein the intermix layer is molecularly integrated with the base material, and wherein the integrated silicon nitride surface layer and the base material cooperate to sandwich the intermix layer in between.
2. The process of claim 1, wherein the ions comprise nitrogen ions selected from the group consisting of N^+ ions or N_2^+ ions.
3. The process of claim 2, wherein the emitting step includes the step of delivering the nitrogen ions at a rate of about 1-5 nitrogen ions for each vaporized silicon atom.
4. The process of claim 3, wherein, when the rate is about one nitrogen ion for each silicon atom, the integrated silicon nitride surface layer includes Si_3N_4 , and, when the rate is about five nitrogen ions for each silicon atom, the integrated silicon nitride surface layer includes SiN_3 .
5. The process of claim 1, including the step of cleaning the outer surface of the orthopedic implant with the ion beam at an energy level between about 1-1000 electron volts.
6. The process of claim 1, wherein the positioning step includes the step of mounting the orthopedic implant to a selectively movable platen for repositioning an orientation of the orthopedic implant relative to the ion beam.
7. The process of claim 1, including the step of vaporizing silicon atoms off a silicon ingot with an evaporator.
8. The process of claim 1, including the step of propagating the ion beam.
9. The process of claim 1, including the step of regulating a formation rate of the silicon nitride molecules by adjusting the ion beam energy or beam density.
10. The process of claim 1, including the step of backfilling the vacuum chamber with the vaporized silicon atoms.
11. The process of claim 1, wherein the integrated silicon nitride surface layer substantially comprises the silicon nitride molecules.
12. The process of claim 1, wherein the driving step includes the step of applying the integrated silicon nitride surface layer to less than an entire outer surface area of the orthopedic implant.
13. The process of claim 1, wherein the integrated silicon nitride surface layer comprises a substantially uniform thickness when driven into the orthopedic implant.
14. A process for producing an orthopedic implant having an integrated silicon nitride surface layer, comprising the steps of: positioning the orthopedic implant inside a vacuum chamber; vaporizing silicon atoms off a silicon ingot with an evaporator; emitting ions via an ion beam into the vaporized silicon atoms in the vacuum chamber to cause a collision between the ions and the vaporized silicon atoms to form reacted silicon nitride molecules; cleaning an outer surface of the orthopedic implant with the ion beam at an energy level between about 1-1000 electron volts; driving the silicon nitride molecules with the same ion beam into the outer surface of the orthopedic implant such that the silicon nitride molecules implant therein and form at least a part of a molecular structure of the outer surface of the orthopedic implant simultaneously while maintaining the outer surface of the orthopedic implant at a temperature below 200 degrees

Celsius, thereby forming the integrated silicon nitride surface layer; and forming an intermix layer underneath the integrated silicon nitride surface layer, the intermix layer including a mixture of subsurface level silicon nitride molecules and a base material of the orthopedic implant, wherein the intermix layer is molecularly integrated with the base material, and wherein the integrated silicon nitride surface layer and the base material cooperate to sandwich the intermix layer in between.

15. The process of claim 14, wherein the ions comprise nitrogen ions selected from the group consisting of N^+ ions or N_2^+ ions and the emitting step includes the step of delivering the nitrogen ions at a rate of about 1-5 nitrogen ions for each vaporized silicon atom.

16. The process of claim 15, wherein, when the rate is about one nitrogen ion for each silicon atom, the integrated silicon nitride surface layer includes Si_3N_4 , and, when the rate is about five nitrogen ions for each silicon atom, the integrated silicon nitride surface layer includes SiN_3 .

17. The process of claim 14, wherein the positioning step includes the step of mounting the orthopedic implant to a selectively movable platen for repositioning an orientation of the orthopedic implant relative to the beam.

18. The process of claim 14, including the step of propagating the ion beam, wherein the integrated silicon nitride surface layer substantially comprises the silicon nitride molecules.

19. The process of claim 14, including the step of regulating a formation rate of the silicon nitride molecules by adjusting the ion beam energy or beam density, wherein the driving step includes the step of applying the integrated silicon nitride surface layer to less than an entire outer surface area of the orthopedic implant.

20. The process of claim 14, including the steps of backfilling the vacuum chamber with the vaporized silicon atoms, wherein the integrated silicon nitride surface layer comprises a substantially uniform thickness when driven into the orthopedic implant.
